Detecting Lies in Video

Shashwat Sharma [1] Enzo Pinchon [1]

[1]Institut Polytechnique de Paris

Introduction - Is this person lying?

Misinformation dominates public discourse, showcased by politicians like Trump. Traditional lie detection methods rely on physiological signals like ECG or brain activity, making them invasive and impractical. AI represents the next step (non invasive, real time detection), and it's why we decided to work on it!

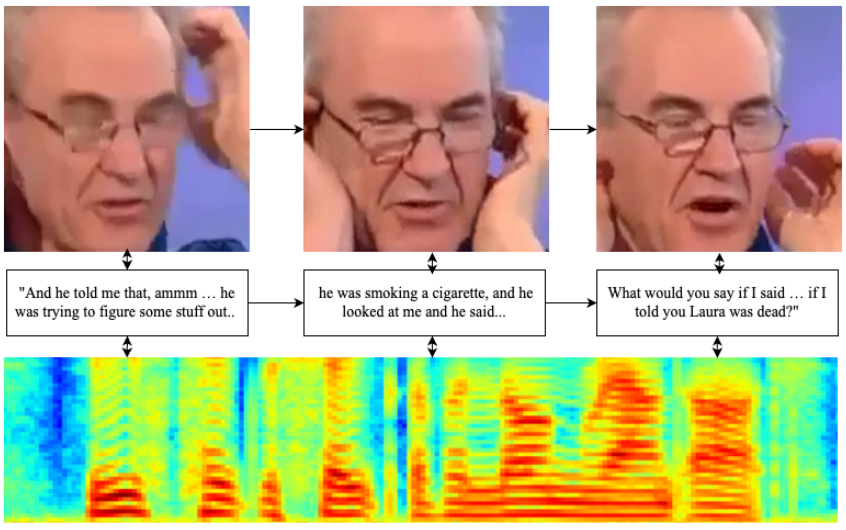


Figure 1. A Training Sample.

Above is part of a data sample that our model processes. Can you identify if the sample is a lie or a truth?

Bibliography

We use the DOLOS dataset which is the Largest gameshow-based deception dataset: 1,675 video clips featuring 213 subjects. Natural conversational deception vs. laboratory-induced scenarios. Rich multimodal data: video, audio with ground truth labels. All participants naturally motivated to deceive (unlike lab settings)

Why Gameshows for Deception Data?

Game shows create high-stakes deception in a controlled setting, perfect for data collection

- **Naturalistic deception:** Participants genuinely motivated to deceive.
- **Clear ground truth:** Definitive knowledge of deceptive vs. truthful statements.
- **Rich conversational context:** Realistic social interaction with stakes.

Baseline Methodology

We employ Parameter-Efficient Crossmodal Learning (PECL) [1] . It integrates Uniform Temporal Adapters (UT-Adapters) within transformer-based encoders for video and audio, ensuring efficient tuning . Additionally, Audio-Visual Fusion module facilitates cross-modal attention, capturing interactions between spoken content and visual cues.

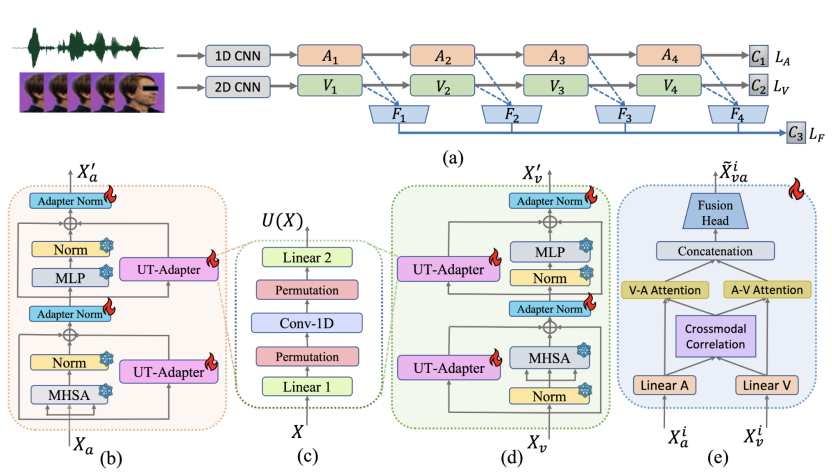


Figure 2. Architectural Overview. [1]

1. **Fine-Tuning:** , UT-Adapters enable lightweight adaptation without full model retraining.
2. **Multimodal Fusion** PAVF captures meaningful interactions between speech and visual expressions.
3. **Multi-task Training:** Jointly learns deception labels and linguistic features for improved performance.

Challenges

Deceptive cues can be subtle temporal patterns across facial expressions, body language, speech, and physiological signals.

- Deceptive behavior **evolve over time** and is context-dependent
- Need for **interpretable models** to explain detection decisions
- **Multiple modalities** need to be fused correctly

We tackle these challenges using an enhanced fused model which can chunk long video sequences and predict lies.

Audio Visual Adapter

The Audio-Visual Fusion module learns crossmodal attention between visual and audio features in an efficient lower-dimensional space. The key is the learnable crossmodal correlation matrix that captures sequence-level interactions:

$$P^i = X_a^i W_P^i X_v^{i\intercal}, \quad \widetilde{X}_v^i = \text{Softmax}(P^i) X_v^i + X_v^i, \quad \widetilde{X}_a^i = \text{Softmax}(P^{i\intercal}) X_a^i + X_a^i$$

Extensions

We integrate Whisper for speech transcription, treating it as an additional modality. Using BERT, we analyze linguistic patterns in deception alongside audio and visual cues. To enhance multimodal learning, we introduce cross-attention for pattern linking and a weighted sum for final fusion.

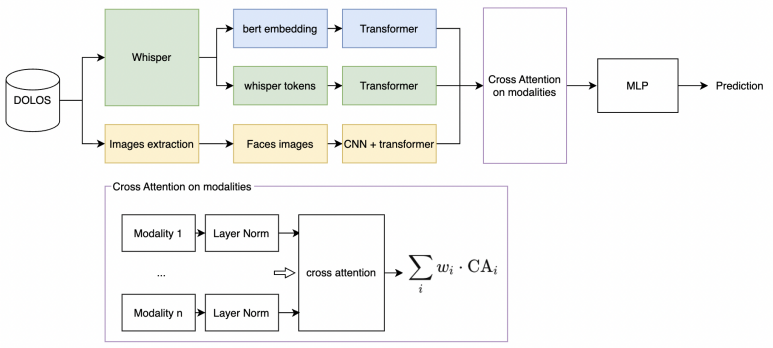


Figure 3. Proposed architecture

Custom Loss Functions

To effectively learn from multiple modalities, we introduce a loss function that balances modality-specific contributions while preserving cross-modal interactions. Our proposed loss combines a contrastive alignment term, modality consistency, and classification objectives.

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{cross-modal}} + \lambda_2 \mathcal{L}_{\text{modality-consistency}} + \lambda_3 \mathcal{L}_{\text{task}}$$

To align representations across different modalities, we employ a contrastive loss inspired by CLIP:

$$\mathcal{L}_{\text{cross-modal}} = -\sum_{i=1}^{N} \log \frac{\exp(\cos(\mathbf{z}_i^A, \mathbf{z}_i^B)/\tau)}{\sum_{j=1}^{N} \exp(\cos(\mathbf{z}_i^A, \mathbf{z}_j^B)/\tau)}$$

To encourage consistency between unimodal and fused representations, we introduce a modality consistency loss:

$$\mathcal{L}_{\text{modality-consistency}} = \sum_{m} \|\mathbf{z}_i^M - \mathbf{z}_i^{\text{fused}}\|_2^2$$

This loss function ensures that our multimodal model learns discriminative and well-aligned representations across audio, video, and text.

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{cross-modal}} + \lambda_2 \mathcal{L}_{\text{modality-consistency}} + \lambda_3 \mathcal{L}_{\text{task}}$$

References

[1] Xiaobao Guo, Nithish Muthuchamy Selvaraj, Zitong Yu, Adams Wai-Kin Kong, Bingquan Shen, and Alex Kot. Audio-visual deception detection: Dolos dataset and parameter-efficient crossmodal learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 22135–22145, 2023.